Expert Systems With Applications 263 (2025) 125728

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

A novel data-driven rolling horizon production planning approach for the plastic industry under the uncertainty of demand and recycling rate

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ARTICLE INFO

Keywords:
Production Planning
Data-driven Optimization
Rolling Horizon Planning
Long Short-Term Memory
Uncertainty
Recycling
Sustainability

ABSTRACT

Efficient production planning in the plastic industry requires integrating sustainability practices, particularly the use of recycled plastics. Recycling helps reduce environmental impacts and conserve resources by decreasing the reliance on virgin materials. To develop a responsive plan, dynamic approaches are beneficial to confront fluctuations in uncertain parameters such as recycling rate and market demand. The present study proposes a novel Data-driven Rolling Horizon Planning (DRHP) approach for a sustainable and dynamic production plan in the plastic industry. A dynamic Rolling Horizon (RH) planning framework is formulated as a multi-product, multi-period Mixed Integer Linear Programming (MILP) model for the problem. This model aims to minimize total production cost while taking into account the system's constraints and sustainability considerations. To deal with uncertainty, the RH-based MILP model is coupled with a rolling Long Short-Term Memory (LSTM) model. The rolling LSTM model leverages historical data of market demand and recycling rate to predict their future values and consequently improve the responsiveness of the production plan. The outperformance of the proposed DRHP approach is demonstrated through an extensive comparison with a robust static counterpart in terms of total production cost and sustainability performance. Results indicate significant reductions, up to 80%, in production costs using the proposed DRHP approach. Furthermore, the effect of rolling duration is investigated concerning production cost, backlog, late-order, and inventory level. Findings highlight the potential of the proposed DRHP approach to mitigate inventory- and late-order-related challenges.

1. Introduction

The use of plastics has expanded across various industries due to their advantageous properties such as durability, ability to resist erosion, versatility, and affordability (Chen et al., 2021; Joseph et al., 2021; Zhang et al., 2021; Tosarkani et al., 2024). As a result, there has been a significant surge in global plastic production over recent decades. The global production of plastics in 2022 reached 400.3 million metric tons, making a 266-fold rise from the 1.5 million metric tons produced in 1950 (Jaganmohan, 2024). This rise presents a pressing challenge for authorities in effectively managing the disposal of a vast amount of nondegradable fossil-based plastic waste.

Approximately 90 % of global plastic production is sourced from fossil-based feedstocks, with packaging representing its primary application at 44 % (Plastics – the facts, 2022). This highlights the importance of employing recycling practices in the packaging industry to mitigate environmental impacts. Plastic packaging comes in different forms (e.g., beverage bottles and food containers) for various purposes. Common types of plastics utilized in the packaging industry include Polyethylene Terephthalate (PET), High-Density Polyethylene (HDPE), Low-Density Polyethylene (LDPE), Polypropylene (PP), Polystyrene (PS), and Polyvinyl Chloride (PVC) (Rabnawaz et al., 2017). Reducing the industry's reliance on virgin plastics through the use of recycled plastics can significantly decrease nondegradable waste in the surrounding ecosystem and conserve natural resources (Shen and Worrell, 2024).

Following the collection, sorting, and cleaning, there are four common standard recycling methods to treat plastic waste: (i) physical recycling, (ii) mechanical recycling, (iii) chemical recycling, and (iv) energy recovery (Faraca et al., 2019; Hahladakis & Iacovidou, 2019; Joseph et al., 2021). These methods are differentiated based on the technology employed and the output of the recycling process. Among the different recycling methods, mechanical recycling stands out as a well-established and industrially feasible approach (Shamsuyeva and Endres, 2021). This method transforms plastic waste into secondary raw materials while maintaining the chemical structure of the original polymer largely intact (Shamsuyeva and Endres, 2021). This process

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https://doi.org/10.1016/j.eswa.2024.125728
Received 26 May 2024; Received in revised form 28 October 2024; Accepted 5 November 2024
Available online 12 November 2024
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involves various stages, including collection, inspection, segregation, sorting, washing, shredding, and extrusion, which can vary to some extent depending on the plastic type and employed technologies (Joseph et al., 2021).

In the mechanical recycling method, material recovery facilities receive collected waste and classify it into distinct types (e.g., PET, HDPE, LDPE, PP, PS, and PVC). These sorted materials are then processed into secondary raw materials such as flakes, pellets, or granules, suitable for manufacturing new products. Depending on factors such as material type, purity, and complexity, as well as recycling technologies, sorting and recycling rates may vary across different facilities (Antonopoulos et al., 2021). Consumption patterns, social awareness, weather conditions, anomalies, events, distance to recycling collection points, and recycling policies are among the other factors influencing the efficiency of recycling programs (Reijonen et al., 2021; Ahmed et al., 2022; Williams and Phillips, 2022). For instance, the plastic waste recycling rate is thirteen times higher when collected separately compared to mixed collection schemes (Plastics – the facts, 2022). In addition to the recycling rate, production plants encounter uncertainty in market demand. Demand fluctuates over time due to various factors such as market trends, consumer preferences, and seasonal variations. Both recycling rate and market demand are potential parameters whose fluctuation can impact predetermined production decisions (e.g., raw material sourcing, production schedules, inventory management, production costs, and sustainability goals).

Most previous studies have tackled production planning problems using static approaches. In a static approach (also known as offline), it is assumed that input data are available prior to planning and the entire scheme is planned all at once in advance (Furini et al., 2015; Lu et al., 2016; Koskinen et al., 2020). As this assumption is not always feasible in the real world (Lu et al., 2023), static approaches are prone to significantly deviate from optimality and result in excessive costs in a real-world setting (Lu et al., 2016). To address this, dynamic approaches are promising solutions as they take into account the changes in input data, acknowledging that the information available to decision-makers may evolve over time (Dehghan Shoorkand et al., 2023; Forel and Grunow, 2023). These approaches incorporate considerations for data uncertainty and can be adjusted in response to updated real-time information (Lu et al., 2016). In this study, a novel dynamic Data-driven Rolling Horizon Planning (DRHP) approach is proposed to optimize production planning, while incorporating recycling opportunities and considering time-varying parameters. To do so, a novel multi-product, multi-period Rolling Horizon (RH)-based Mixed-Integer Linear Programming (MILP) model is developed. The aim is to minimize total production costs, including costs of raw material sourcing, recycling activities, production activities, and inventory holding, as well as unfulfilled demand, backlogs, and late-order. A Long Short-Term Memory (LSTM) model is coupled with the RH-based model to review and update the predictions of uncertain parameters throughout the planning scheme. As updating frequency is a key parameter affecting prediction accuracy and the RH-based model's outcome, a comprehensive analysis is also conducted to investigate its impacts. Furthermore, the performance of the developed DRHP approach is evaluated against its robust static counterpart in terms of total production cost and sustainability considerations. The main contributions of this research can be summarized as follows:

- Developing a dynamic RH-based optimization framework integrated with a rolling LSTM model for sustainable production planning in the plastic industry, capable of timely response to changes in recycling rate and demand.
- Demonstrating the superior performance of the proposed DRHP approach over its static counterpart in terms of total production cost and sustainability performance under different scenarios of rolling duration.
- Investigating the impact of updating frequency, also known as rolling duration, on total production cost, raw material and final product inventories, and raw material shortage.

The remainder of this study is organized as follows: Section 2 provides a literature review on the applications of RH approaches in time-dependent optimization problems, along with recent studies in data-driven production planning. Section 3 explains the developed RH-based mathematical model and the proposed DRHP approach as well as its robust static counterpart. The performance of the proposed approach is evaluated across different instances and results are discussed in Section 4. Finally, Section 5 summarizes the main findings of the study, discusses the limitations, and offers suggestions for future research directions.

2. Literature review

In this section, a comprehensive literature review is conducted to explore various aspects of the study, including RH planning, demand forecasting, and data-driven optimization. Furthermore, the research gaps and challenges of the topic are discussed in detail.

2.1. RH planning

Today, most optimization problems are formulated over a set of time periods. To tackle optimization problems having a time dimension, RH approaches have been widely used in different areas such as production planning (Clark, 2005; Li and Ierapetritou, 2010; Ziarnetzky et al., 2018; Koskinen et al., 2020; Wu et al., 2021; Torkaman et al., 2022), transportation planning (Furini et al., 2015; Cavagnini et al., 2022; Zaneti et al., 2022; Lu et al., 2023), energy planning (Marquant et al., 2015; Cuisinier et al., 2022; Shalaby et al., 2022; Abedi and Kwon, 2023), and supply chain management (Lu et al., 2016; Fattahi and Govindan, 2022). RH approaches decompose the computationally expensive main problem into smaller subproblems that can be solved within a reasonable time. Glomb et al. (2022) introduced a framework to apply an RH approach in various large-scale optimization problems with a time dimension. They outlined conditions under which this framework could guarantee the optimality of the generated solutions. In addition to time-dependent problems, RH approaches have been proven beneficial in dynamic and uncertain environments, where parameters may be subject to changes (Ziarnetzky et al., 2018; Fattahi and Govindan, 2022; Shalaby et al., 2022; Abedi and Kwon, 2023; Dehghan Shoorkand et al., 2023; Forel and Grunow, 2023).

Given the dynamic nature of production environments, RH approaches can effectively address multi-period production planning problems. Tolio and Urgo (2007) developed an RH-based model to address the challenges of production planning and outsourcing under uncertainty. They incorporated the uncertain nature of resource needs by formulating the problem within a scenario-based framework. Their proposed RH approach was based on a two-stage stochastic programming method, accounting for multiple uncertain scenarios. The results highlighted the advantages of their proposed model over planning approaches based only on the expected values of uncertain variables. Li and Ierapetritou (2010) leveraged an RH approach to incorporate production capacity information into the production planning and scheduling problem. The findings showed that a precise capacity model led to near-optimal solutions. This was credited to the model's capability to forecast future costs associated with potential backorders when informed by production capacity data. Torkaman et al. (2022) studied the problem of production-routing with price-dependent demand within an outer approximation framework. They solved the primal problem using a non-linear programming solver, but to tackle the mixed integer programing master problem, they developed four different RH-based heuristics. They showed the outperformance of their proposed heuristics in addressing the computationally expensive master problem compared to other algorithms, with minimal or no deviation from the optimal solution. Ziarnetzky et al. (2018) showed the improved performance of a chance-constrained water fab production planning model incorporating demand forecast updates. They employed a multiplicative martingale model of forecast evolution and illustrated the outperformance of a RH-based model in terms of expected service level and profit.

Forel and Grunow (2023) developed a dynamic stochastic, capacitated lot-sizing methodology adapted to RH planning. They integrated forecast evolution models into lot-sizing formulations, efficiently incorporating cumulative demand distributions using linearization techniques. The authors proved the effectiveness of the proposed methodology through a large-scale numerical analysis with real-world and synthetic data. The results demonstrated a significant cost reduction compared to traditional deterministic and stochastic methods relying only on demand history. Dehghan Shoorkand et al. (2023) considered data-driven predictive maintenance in a multi-period, multi-product capacitated lot-sizing problem based on an RH approach. They integrated an LSTM model with an RH-based model to dynamically forecast and update the health conditions of machines along with tactical production decisions. The proposed integrated approach was validated on a set of benchmarks, and its outperformance over other solution approaches was shown. In addition to production planning, the integration of data-driven models with RH approaches has attracted attention in other domains. Abedi and Kwon (2023) developed an RH-based optimization model aimed at improving the efficiency of battery energy storage operations. They coupled the RH-based model with an LSTM model to address the uncertainty of demand, price, and solar power generation. The proposed integrated model iteratively forecasted uncertainty and optimized operational decisions. Evaluation against an offline model demonstrated the satisfactory performance of their proposed model.

2.2. Data-driven production planning optimization

Data-driven optimization has recently attracted great interest across various industries and academic research. Data-driven optimization approaches widely employ machine learning and other artificial intelligence techniques, often coupled with mathematical optimization models, to optimize processes and systems. Data-driven optimization enables decision-makers to address the challenges of coordinating various elements of an optimization problem in dynamic environments by integrating medium-term planning and short-term scheduling (Ziarnetzky et al., 2018; Hubbs et al., 2020; Santander et al., 2020; Beykal et al., 2022; Lee and Lee, 2022; Vieira et al., 2022; Dehghan Shoorkand et al., 2023). In the field of production planning, Hubbs et al. (2020) proposed a Reinforcement Learning (RL) model to dynamically schedule a single-stage multi-product chemical process. Their study demonstrated the superior performance of the approach compared to conventional scheduling based on a receding horizon MILP model. The authors argued that handling uncertainty through simulation techniques was more advantageous than within a mathematical programming model. Beykal et al. (2022) developed a data-driven optimization approach for the integrated problem of production planning and scheduling under uncertain demand. They extended a mixed-integer bilevel multi-follower programming, based on the DOMINO algorithm, and conducted stochastic analysis of various product demand scenarios to provide feasible solutions for complex industrial processes. Lee and Lee (2022) employed a Deep Reinforcement Learning (DRL) algorithm (i.e., deep Q-network) to optimize production scheduling. They aimed to ensure a realistic decision-making process considering the complex re-entry characteristics and long lead times in the semiconductor industry. They showed that by a novel state-action-reward formulation, the DRL method outperforms other dispatching rules, offering robust solutions in dynamic manufacturing environments with high uncertainty. Vieira et al. (2022) proposed a recursive optimization-simulation approach to facilitate decision-making for efficient production planning and scheduling in a human-robot collaborative environment. This approach determined plans through a two-level MILP model and iteratively evaluated them by a discrete-event simulation model to guarantee capacity-feasible solutions at the scheduling level..

2.3. Demand forecasting

Production planners should determine daily production quantity to meet market demand at minimum production costs, while also considering system constraints. Market demand is a key parameter in production planning problems, given its potential to significantly impact production scheduling, raw material sourcing, and inventory-related decisions. Given the case study of PET bottles in this study, it is beneficial to explore forecasting models used to predict daily urban water consumption.

When dealing with demand forecasting, statistical methods stand out as efficient tools. Du et al. (2020) proposed an auto-regressive integrated moving average model combined with a Markov chain for water consumption prediction. This study showed that the prediction error of their proposed model can be considerably reduced by using the Markov model for predicting data trends. With the growing applications of Artificial Neural Network (ANN)-based methods in predictive analysis, they have emerged as a popular choice for demand forecasting across diverse industries and domains. Farah et al. (2019) employed a feed-forward back-propagation ANN model to forecast water consumption on the campus of Lille University. The results highlighted the acceptable performance of the ANN model on both daily and hourly analyses, including peak values. Benítez et al. (2019) indicated that good prediction accuracy can be heavily challenged by the inclusion of various seasonality patterns. In this regard, they adopted a pattern-similarity method, removing the need for annual seasonality determination and extending the horizon of prediction. Kühnert et al. (2021) reported the outperformance of LSTM models over other conventional methods that previously employed. They showed that LSTM models can capture and integrate the effect of seasonality and events on water demand while only requiring a few days of training data. To address the inherent uncertainty in water demand, Du et al. (2022) developed a hybrid model that combines LSTM networks with Kernel Density Estimation (KDE). They optimized their proposed model by employing the Particle Swarm Optimization (PSO) algorithm. The authors utilized the hybrid KDE-PSO-LSTM for predicting intervals of urban water demand, incorporating a novel splitting strategy to classify prediction errors into different levels. By optimizing bandwidth parameters and applying a confidence-window shifting method, the model aimed to accurately quantify uncertainties in predictions and facilitate water supply planning.

Pu et al. (2023) integrated wavelet multi-resolution analysis into a Convolutional Neural Network (CNN)-LSTM model for short-term water demand forecasting. Their objective was to assist water utilities in reducing energy costs and mitigating unforeseen disruptions. They demonstrated the superior performance of the hybrid wavelet-CNN-LSTM model compared to previously used models in the literature, achieving enhanced accuracy in both single- and multi-step predictions. Iwakin and Moazeni (2024) proposed a probabilistic forecasting method for predicting future water consumption patterns, considering the inherent uncertainties in the system. By leveraging statistical techniques, they developed a conformal prediction-based hybrid demand forecast model, incorporating CNN and bidirectional LSTM. Validation in real-world settings demonstrated the effectiveness of the proposed framework in generating accurate deterministic demand forecasts and reliable adaptive probability estimates. Overall, the literature demonstrates the potential advantages of ANN-based methods for enhancing decision-making and resource management within water distribution systems.

2.4. Research gaps

Table 1 presents a comparative analysis of recent studies on the production planning and scheduling problem by highlighting their different characteristics. A thorough review demonstrates a research gap in the literature concerning dynamic production planning in the plastic industry that incorporates recycling plastics as an alternative for virgin plastics and considers uncertainty challenges of recycling rate and market demand. Furthermore, the important impact of updating strategy on the planning outcomes was not discussed in the previous studies. To address the research gap, the present study develops a novel dynamic multi-product, multi-period RH-based optimization model tailored for production planning in the plastic industry considering recycling opportunities. To tackle uncertainty, the proposed mathematical model is augmented with a rolling LSTM model to forecast future values of market demand and recycling rate and use them for informed decision-making. Additionally, the study conducts a sensitivity analysis on rolling duration, a key parameter that directly affects updating strategy.

3. Problem statement and methodology

3.1. Problem description

Production planning is defined as a series of decisions concerning production scheduling, raw material sourcing, and inventory management (Bueno et al., 2020). Unlike conventional planning, which adheres to predetermined decisions for the entire planning scheme, the present study introduces a dynamic framework that allows for the integration of updated data to adjust production planning accordingly. The proposed framework is inspired by Abedi and Kwon (2023) and Dehghan Shoorkand et al. (2023).

In this study, demand and recycling rate are considered uncertain parameters whose actual values are gradually revealed over time. Once a raw material order is placed, whether through virgin plastics stream or recycled plastics stream, it is considered irrevocable. In a case where the recycling stream is selected, the quantity of the required plastic waste is calculated based on the forecasted recycling rate. Upon receiving the plastic waste and determining its actual recycling rate, the actual-added raw material is updated. This value is replaced with its previously forecasted value and the production plan is adjusted accordingly. The reason is that the raw material inventory level affects inventory holding costs and subsequent production decisions. In addition to the recycling rate, future demand is another determinant influencing the production systems in terms of production scheduling and raw material sourcing. This is due to the fact that market demand can noticeably influence the backlog and the inventory level. In the proposed approach, the production plan is updated by evolving the available information on actual market demand as well. Having said this, the accuracy of recycling rate and market demand predictions is crucial, especially considering penalties for raw material late-order, unfulfilled demand, and production capacity constraints.

The primary objective of this study is to demonstrate the advantages of the proposed dynamic data-driven optimization approach to decrease production costs and improve sustainability performance. This approach integrates both short- and medium-term decisions without compromising the effectiveness of either.

3.2. The RH-based model

RH planning is an adaptive approach that involves updating a plan over time as new information becomes available (Cuisinier et al., 2022; Fattahi and Govindan, 2022; Glomb et al., 2022; Abedi and Kwon, 2023; Dehghan Shoorkand et al., 2023; Forel and Grunow, 2023). The main idea is to break a longer-term planning problem into a series of shorter-term planning horizons and to update the plan as new information becomes available (Tolio and Urgo, 2007; Li and Ierapetritou, 2010). In an RH approach, a plan is initially created for the first horizon, typically covering a short period of time. As the first horizon comes to an end, the plan is adjusted for the next horizon, taking into account the current system status, updated information, and forecasts of the future. This iterative process is repeated for subsequent horizons as well. This approach enables planners to respond quickly to changes in the environment (e.g., unexpected events or changes in market conditions), and to make more accurate and informed decisions.

In this study, the production planning problem is defined under H horizons. It is assumed that each horizon constitutes Θ operational periods and Ξ forecasting periods ($\Theta \leq \Xi$), each period spanning τ days. The parameters Θ and Ξ are introduced to define the durations of operational and forecasting sets, with the assumption that raw material delivery/preparation takes $\Theta\tau$ days. At the beginning of each horizon, the actual demands for operational periods (the first $\Theta\tau$ days) are available, while forecasting periods remain uncertain. The developed production plan of each horizon is mandatory to be implemented within the operational periods, yet it can be further modified for the forecasting periods by moving to the subsequent horizons. Regular raw material orders, whether recycled or virgin plastics, are placed at the beginning of the operational set and are received at the beginning of the forecasting set of the same horizon. The order for the plastic waste stream (if decided) is placed based on a predicted recycling rate, while the actual-added raw material is further revealed based on the actual recycling rate. Raw material late-order aims to compensate for raw material shortage resulting from overestimating the recycling rate in the previous horizon. Late-order is placed and received at the beginning of the operational set in each horizon. By moving to the next horizon (shifting $\Theta\tau$ days forward), new operational and forecasting sets are defined, and the optimization process is repeated. The developed RH-based MILP model is defined as follows for each realization of the horizon h.

Indices:

t	Days; $t = (h-1)\Theta\tau + 1, \ldots, (h-1)\Theta\tau + (\Theta + \Xi)\tau$.
p	Type of raw materials/products; $p = 1, \cdots, P$.

Parameters:

D_{pt}^{h}	Demand of product p on day t during operational periods of horizon h (unit).
D'^{h}_{pt}	Predicted demand of product p on day t during forecasting periods of horizon h (unit).
RR'^{h}_{p}	Predicted recycling rate of the plastic waste type p in horizon h (percent).
L_{p}^{h}	Actual amount of raw material type p sourced through recycling stream for horizon h (kg).
X_{p}^{h}	Actual amount of raw material type p sourced through virgin plastics stream for horizon h (kg).
$Pcap_{pt}^{h}$	Production capacity for product type p on day t in horizon h (unit).
$Cinvr_{pt}^{h}$	Inventory cost of raw material type p on day t in horizon h ($ per kg).
$Cinvf_{pt}^{h}$	Inventory cost of final product type p on day t in horizon h ($ per unit).
Cdo_{p}^{h}	Late-order cost for virgin plastic type p in horizon h ($ per kg).
$Cpro_{pt}^{h}$	Production cost of product type p on day t in horizon h ($ per unit).
Cbl_{pt}^{h}	Backlog cost of product type p on day t in horizon h ($ per unit).
Cvp_{p}^{h}	Cost of virgin plastic type p in horizon h ($ per kg).
Cwp_{p}^{h}	Cost of plastic waste type p in horizon h ($ per kg).
Cr_{p}^{h}	Cost of recycling plastic waste type p in horizon h ($ per kg).
Ccv_{p}^{h}	Contract cost for virgin plastic type p in horizon h ($).
Ccw_{p}^{h}	Contract cost for plastic waste type p in horizon h ($).
$Invr_{p}^{h}$	Inventory of raw material type p at the beginning of horizon h (kg).
$Invf_{p}^{h}$	Inventory of final product type p at the beginning of horizon h (unit).
α_{p}	Unitary raw material consumption for product type p (percent).
M	A very large positive number.
ε	A very small positive number.

Table 1
A summary of the recent literature on production planning and scheduling.

Authors	Industry	Mathematical optimization models	Multi-period programming	Data-driven techniques	Sustainability considerations	Uncertain environment	Dynamic environment	Analysis of updating policy	Methodology
Ziarnetzky et al. (2018)	Semiconductor fabrication	✓	✓	–	–	✓	✓	–	Rolling horizon planning
Rossit et al. (2019)	Cyber-physical production system	–	–	✓	–	✓	✓	–	Big data techniques
Kamal et al. (2019)	–	✓	–	–	✓(Social)	✓	–	–	Fuzzy goal programming
Hubbs et al. (2020)	Chemical production process	✓	✓	✓	–	✓	✓	–	Deep reinforcement learning and receding horizon planning
Kim et al. (2020)	Construction industry	–	✓	–	–	✓	✓	–	Discrete-time simulation
Mohammadi et al. (2020)	Furniture manufacturing	✓	–	–	–	–	–	–	Hybrid particle swarm optimization algorithm
Morariu et al. (2020)	–	–	✓	✓	–	✓	✓	–	Big data and machine learning
González Rodríguez et al. (2020)	Industrial hospital laundry	–	–	✓	–	✓	✓	–	Fuzzy logic and machine learning
Santander et al. (2020)	Chemical manufacturing	✓	✓	–	–	✓	✓	–	–
Sehovac and Grolinger (2020)	Electricity planning	–	✓	✓	✓ (Environmental)	✓	✓	–	Attention mechanism and long short-term memory
Goli et al. (2021)	–	✓	–	–	–	✓	–	–	A hybrid of genetic algorithm and wale optimization algorithm
Guo et al. (2021)	–	✓	–	–	✓ (Environmental)	✓	–	–	Stochastic hybrid discrete grey wolf optimizer
Villalonga et al. (2021)	Cyber-physical production system	–	–	✓	–	✓	✓	–	Fuzzy logic and machine learning
Beykal et al. (2022)	–	✓	✓	✓	–	✓	–	–	DOMINO algorithm
Gahm et al. (2022)	Metal processing	–	–	✓	–	✓	–	–	Machine learning algorithms
Lee and Lee (2022)	Semiconductor fabrication	✓	✓	✓	–	✓	✓	–	Deep reinforcementlearning
Vieira et al. (2022)	–	✓	✓	–	–	✓	✓	–	Recursive optimization-simulation approach
Zhang et al. (2022a)	–	–	✓	✓	–	✓	✓	–	Deep reinforcementlearning
Zhang et al. (2022b)	–	✓	–	–	–	–	–	–	Hierarchical multi-strategy genetic algorithm
Dehghan Shoorkand et al. (2023)	–	✓	✓	✓	–	✓	✓	–	Rolling horizon planning
Ghaedy-Heidary et al. (2023)	Semiconductor manufacturing	✓	–	–	–	✓	–	–	Simulation optimization, and genetic algorithm
Hameed and Schwung (2023)	Injection molding	–	–	✓	–	–	✓	–	Tabu search, genetic algorithm, and reinforcement learning
Yağmur and Kesen (2023)	–	✓	–	–	✓ (Environmental)	–	–	–	Partial local search and non-dominated sorting genetic algorithm-ii
Zhang et al. (2023)	–	✓	–	–	–	–	–	–	Improved artificial bee colony algorithm
Tosarkani et al. (2024)	Plastic industry	✓	✓	✓	✓ (Environmental and social)	✓	–	–	Multi-objective data-driven stochastic

(continued on next page)

Table 1 (*continued*)

Authors	Industry	Mathematical optimization models	Multi-period programming	Data-driven techniques	Sustainability considerations	Uncertain environment	Dynamic environment	Analysis of updating policy	Methodology
Current study	Plastic industry	✓	✓	✓	✓ (Environmental)	✓	✓	✓	possibilistic programming Data-driven rolling horizon planning

Decision variables:

k^h_p	Amount of late-order raw material type p in horizon h (kg).
l'^h_p	Amount of raw material type p expected to be sourced through recycling stream for forecasting periods in horizon h (kg).
pq^h_{pt}	Number of product type p produced on day t in horizon h (unit).
sq^h_{pt}	Number of product type p sold on day t in horizon h (unit).
$invr^h_{pt}$	Inventory of raw material type p at the end of day t in horizon h (kg).
$invf^h_{pt}$	Inventory of final product type p at the end of day t in horizon h (kg).
v'^h_p	Equals 1 if virgin plastic type p contributes to raw material sourcing for forecasting periods in horizon h, otherwise 0.
w'^h_p	Equals 1 if recycling stream contributes to raw material sourcing of product type p for forecasting periods in horizon h, otherwise 0.
x'^h_p	Amount of virgin plastic type p ordered for forecasting periods in horizon h (kg).
y'^h_p	Amount of plastic waste type p ordered for forecasting periods in horizon h (kg).

According to Lohmer and Lasch, (2021), cost minimization is one of the most popular objective functions in the production planning and scheduling literature. This helps decision-makers develop a plan so as to prevent excessive costs while considering market demand and production constraints. In this study, the total production cost in horizon h (i.e., Z_h) is defined as the summation of different cost elements (i.e., Z_{h1}, Z_{h2}, Z_{h3}, and Z_{h4}). Z_{h1} accounts for the total raw material sourcing costs, either through virgin or recycled plastics. Z_{h2} calculates the costs of inventory holding and production activities throughout the entire horizon h. Finally, Z_{h3} and Z_{h4} take into account the backlog costs during operational and forecasting sets, respectively. The cost components considered in Z_{h2}, Z_{h3}, and Z_{h4} are inspired by Mishra et al. (2020) and Dehghan Shoorkand et al. (2023).

$$Z_{h1} = \sum_{p=1}^{P}\left(Cdo^h_p.k^h_p + Cvp^h_p.x'^h_p + Cwp^h_p.y'^h_p + Cr^h_p.l'^h_p + Ccv^h_p.v'^h_p + Ccw^h_p.w'^h_p\right) \tag{1-1}$$

$$Z_{h2} = \sum_{t=(h-1)\Theta\tau+1}^{(h-1)\Theta\tau+(\Theta+\Xi)\tau}\sum_{p=1}^{P}\left(Cinvr^h_{pt}.invr^h_{pt} + Cinvf^h_{pt}.invf^h_{pt} + Cpro^h_{pt}.pq^h_{pt}\right) \tag{1-2}$$

$$Z_{h3} = \sum_{t=(h-1)\Theta\tau+1}^{(h-1)\Theta\tau+\Theta\tau}\sum_{p=1}^{P} Cbl^h_{pt}.\left(D^h_{pt} - sq^h_{pt}\right) \tag{1-3}$$

$$Z_{h4} = \sum_{t=(h-1)\Theta\tau+\Theta\tau+1}^{(h-1)\Theta\tau+(\Theta+\Xi)\tau}\sum_{p=1}^{P} Cbl^h_{pt}.\left(D'^h_{pt} - sq^h_{pt}\right) \tag{1-4}$$

Objective function:

$$MinZ_h = Z_{h1} + Z_{h2} + Z_{h3} + Z_{h4} \tag{1}$$

Constraints:

$$pq^h_{pt} \le Pcap^h_{pt};\forall t\in\{(h-1)\Theta\tau+1,\cdots,(h-1)\Theta\tau+(\Theta+\Xi)\tau\},\forall p\in\{1,\cdots,P\} \tag{2}$$

$$sq^h_{pt} \le D^h_{pt};\ \forall t \in \{(h-1)\Theta\tau+1,\cdots,(h-1)\Theta\tau+\Theta\tau\},\ \forall p \in \{1,\cdots,P\} \tag{3}$$

$$sq^h_{pt} \le D'^h_{pt};\forall t\in\{(h-1)\Theta\tau+\Theta\tau+1,\cdots,(h-1)\Theta\tau+(\Theta+\Xi)\tau\},\forall p\in\{1,\cdots,P\} \tag{4}$$

$$Invr^h_p + X^h_p + L^h_p + k^h_p - \alpha_p pq^h_{pt} = invr^h_{pt};\ t = (h-1)\Theta\tau+1,\ \forall p \in \{1,\cdots,P\} \tag{5}$$

$$invr^h_{p(t-1)} - \alpha_p pq^h_{pt} = invr^h_{pt};\ \forall t \in \{(h-1)\Theta\tau+2,\cdots,(h-1)\Theta\tau+(\Theta+\Xi)\tau\} - ((h-1)\Theta\tau+\Theta\tau+1),\ \forall p \in \{1,\cdots,P\} \tag{6}$$

$$invr^h_{p(t-1)} + x'^h_p + l'^h_p - \alpha_p pq^h_{pt} = invr^h_{pt};\ t = (h-1)\Theta\tau+\Theta\tau+1,\ \forall p \in \{1,\cdots,P\} \tag{7}$$

$$Invf^h_p + pq^h_{pt} - sq^h_{pt} = invf^h_{pt};\ t = (h-1)\Theta\tau+1,\ \forall p \in \{1,\cdots,P\} \tag{8}$$

$$invf^h_{p(t-1)} + pq^h_{pt} - sq^h_{pt} = invf^h_{pt};\ \forall t \in \{(h-1)\Theta\tau+2,\cdots,(h-1)\Theta\tau+(\Theta+\Xi)\tau\},\ \forall p \in \{1,\cdots,P\} \tag{9}$$

$$\varepsilon.v'^h_p \le x'^h_p \le M.v'^h_p;\ \forall p \in \{1,\cdots,P\} \tag{10}$$

$$\varepsilon.w'^h_p \le y'^h_p \le M.w'^h_p;\ \forall p \in \{1,\cdots,P\} \tag{11}$$

$$RR'^h_p.y'^h_p = l'^h_p;\ \forall p \in \{1,\cdots,P\} \tag{12}$$

$$k_p^h, l'^h_p, pq^h_{pt}, sq^h_{pt}, invr^h_{pt}, invf^h_{pt}, x'^h_p, y'^h_p \in R^+;\ \forall t \in \{(h-1)\Theta\tau+1, \cdots, (h-1)\Theta\tau+(\Theta+\Xi)\tau\},\ \forall p \in \{1, \cdots, P\} \tag{13}$$

$$v'^h_p, w'^h_p \in \{0,1\}; \forall t \in \{(h-1)\Theta\tau+1, \cdots, (h-1)\Theta\tau+(\Theta+\Xi)\tau\}, \forall p \in \{1, \cdots, P\} \tag{14}$$

In the developed MILP model, Objective (1) minimizes the expected total production cost in horizon h. This is formulated based on costs related to raw material sourcing, recycling activities, production activities, inventory handling, backlog, and late-order. Constraint (2) takes production capacity limits into account. Constraints (3) and (4) consider the impact of market demand on selling quantity. Constraints (5)–(7) track the flow of raw materials inventory throughout the entire horizon h. Constraints (8) and (9) balance the inventory flow of final products. Constraints (10) and (11) determine through which streams raw materials are sourced. Constraint (12) determines the expected amount of raw materials sourced from the recycling stream. Constraints (13) and (14) specify the domain of the decision variables.

In this model, $L_p^h, X_p^h, Invr_p^h, Invf_p^h$ are calculated based on the outputs of the previous horizon and recently revealed data where $L_p^h = RR_p^{h-1}.y'^{h-1}_p$, $X_p^h = x'^{h-1}_p$ for $p \in \{1, \cdots, P\}$ and $Invr_p^h = invr_{pt}^{h-1}$, $Invf_p^h = invf_{pt}^{h-1}$ for $p \in \{1, \cdots, P\}$ and $t = (h-1)\Theta\tau$.

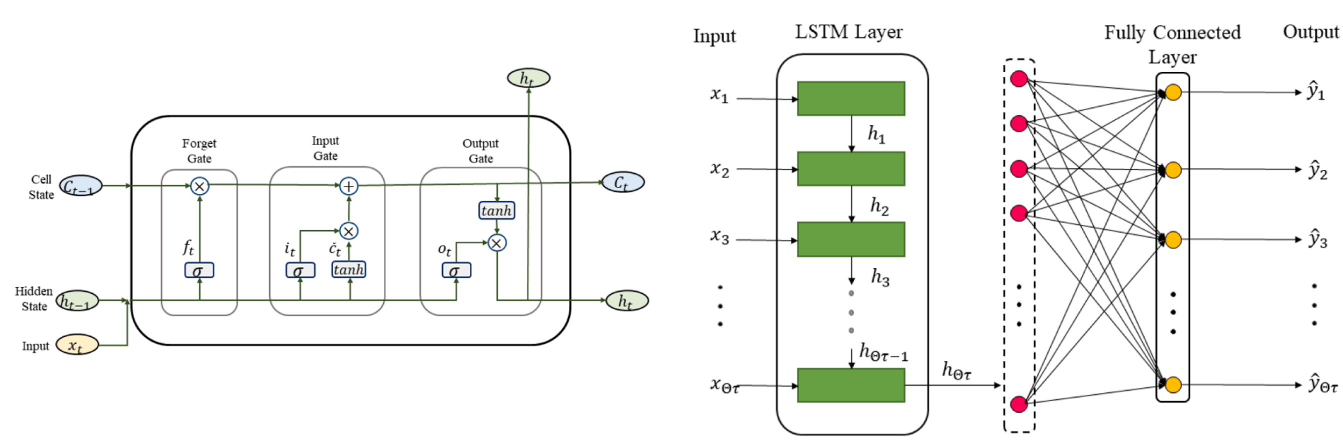
After rolling through all the horizons, the Actual Total Cost (ATC) is calculated for the entire planning scheme as follows:

$$ATC := \sum_{h=1}^{H}\sum_{p=1}^{P}\left(Cdo_p^h.k_p^h + Cvp_p^h.x'^h_p + Cwp_p^h.y'^h_p + Cr_p^h.L_p^{h+1} + Ccv_p^h.v'^h_p + Ccw_p^h.w'^h_p\right) + \sum_{t=(h-1)\Theta\tau+1}^{(h-1)\Theta\tau+\Theta\tau}\sum_{p=1}^{P}\left(Cinvr_{pt}^h.invr_{pt}^h + Cinvf_{pt}^h.invf_{pt}^h + Cpro_{pt}^h.pq_{pt}^h + Cbl_{pt}^h.\left(D_{pt}^h - sq_{pt}^h\right)\right) \tag{15}$$

3.3. The rolling LSTM model

In this section, a rolling LSTM model (compatible with the proposed RH-based model) is presented to predict future values of uncertain parameters (i.e., RR_p^h and D'^h_{pt}). LSTM models are a specific category of Recurrent Neural Networks (RNNs) designed to address the vanishing/exploding gradient problem that occurs when modelling long-term dependencies in time-series data. In standard RNNs, gradients can either diminish or escalate significantly during backpropagation through multiple layers, hindering the network's ability to effectively learn long-term relationships within the data. An LSTM model tackles this challenge by introducing a cell state and three different gates at each recurrent unit: (i) input gates for adding information to the cell, (ii) forget gates for discarding unnecessary information, and (iii) output gates for selecting and outputting relevant information. The cell state, akin to memory, allows the LSTM model to retain and manipulate information over extended sequences, effectively capturing temporal dependencies and enabling accurate predictions.

LSTM models have proven advantageous in forecasting water consumption profiles, as understanding long-term patterns, seasonal variations, and behavioural trends is essential for accurate demand forecasting (Kühnert et al., 2021). Additionally, they have shown to be effective in predicting recycling rate and improving resource utilization



a) b)

Fig. 1. Architecture of a) an LSTM cell, and b) the employed LSTM model.

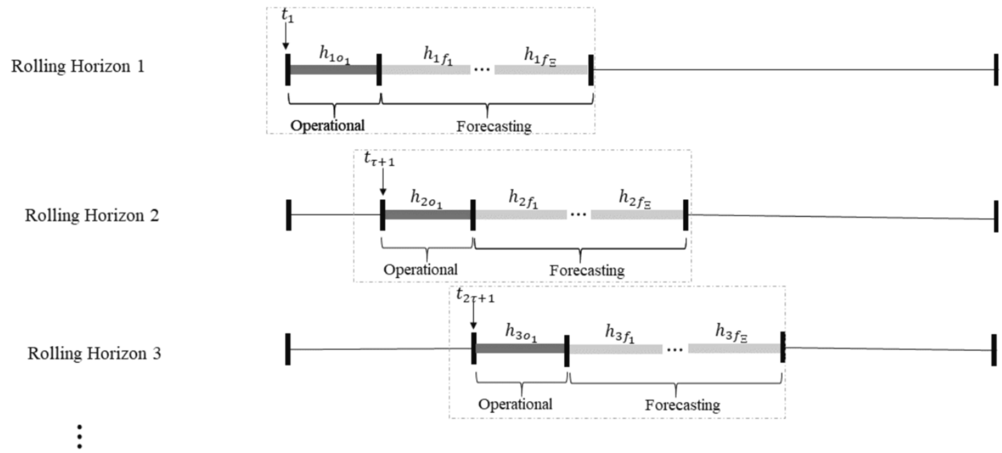


Fig. 2. An example of the proposed RH planning framework ($\Theta = 1$).

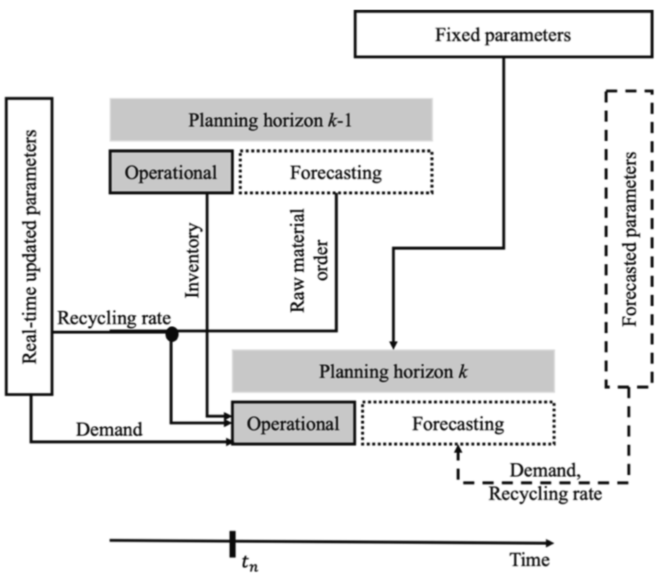


Fig. 3. Flow of parameters in the k^{th} planning horizon.

in waste management systems (Olawumi et al., 2024). The ability of LSTM models to effectively handle long-term dependencies makes them favourable for solving intricate time-series prediction tasks. They are helpful to model complex nonlinear relationships and derive insightful patterns from sequential observations. Fig. 1 illustrates the classic representation of an LSTM cell (Nasser et al., 2020) and the LSTM model used in this study.

The notations in Fig. 1.a) are explained by the following equations:

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{16}$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{17}$$

$$\widetilde{c}_t = tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{18}$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{19}$$

$$c_t = f_t \otimes c_{t-1} + i_t \otimes \widetilde{c}_t \tag{20}$$

$$h_t = o_t \otimes \tanh(c_t) \tag{21}$$

where h_{t-1} represents the previous hidden state and x_t denotes the current time step's input. In the presented cell state, f_t, i_t, o_t correspond to the forget gate, input gate, and output gate, respectively. $\widetilde{c}_t$ represents the candidate cell state used for updating the cell state. c_t is the current cell state, c_{t-1} is the previous cell state, and h_t represents the current hidden state. The weighted matrices are denoted by W_f, W_i, W_c, W_o and U_f, U_i, U_c, U_o while b_f, b_i, b_c, b_o represent bias terms. All these parameters are trainable. The symbol $\otimes$ signifies the Hadamard product; σ denotes the sigmoid function, and tanh represents the TanHyperbolic function.

Algorithm 1. Outlines the rolling LSTM model employed in this study. To train neural networks effectively, learning algorithms (e.g., back-propagation) enhanced with adaptive step sizes are widely used. In this study, the ADAM optimizer is employed to improve the performance of the forecasting model. ADAM optimizer is a popular gradient-based optimization algorithm that adaptively adjusts learning rates for each parameter during training and is known for fast convergence (Bock et al., 2018). The present rolling LSTM model is designed so as to receive the actual data of the previous $\Theta\tau$ days (i.e., $X_i = (x_1, \cdots, x_{\Theta\tau})$) and predicts the data for the subsequent $\Theta\tau$ days (i.e., $\widehat{Y}_i = (\widehat{y}_1, \cdots, \widehat{y}_{\Theta\tau})$). Algorithm 1 is integrated into the DRHP approach as a predicting tool to improve the performance of the RH-based MILP model (refer to Algorithm 2).

Algorithm 1. *Rolling LSTM Model.*

Input: Historical data, train-test ratio, τ and Θ.
Step 1: Normalize data; $x_i = \dfrac{x_i - x_{min}}{x_{max} - x_{min}}$
Step2: Determine rolling duration $= \Theta\tau$
Step 3: Create sequential data
for $i = 0, 1, 2, \cdots, |Historicaldata| - 2\Theta\tau$ do
    i_th observation: $X_i \leftarrow (x_{i+1}, \cdots, x_{i+\Theta\tau})$
    i_th label: $Y_i \leftarrow (x_{i+\Theta\tau+1}, \cdots, x_{i+2\Theta\tau})$
    return X_i and Y_i
end for
Step 4: Shuffle sequential data
Step 5: Split sequential data into train and test sets based on the train-test ratio
$M \rightarrow \{(X_i, Y_i); \forall i \in \textit{train set}\}$
$N \rightarrow \{(X_i, Y_i); \forall i \in \textit{test set}\}$
Step 6: Model = sequential ()
Step 7: Model.add (LSTM (units))
Step 8: Model.add (Dense($\Theta\tau$))
Step 9: Model.compile (Loss = RMSE, Optimizer = ADAM)
Step 10: ***For*** epoch = 1, 2, …, do
    Model.fit (X_M, Y_M, epochs)
    Return Model
End for
Step 11: Model.predict (X_M)
Step 12: Model.predict (X_N)
Step 13: Report RMSE values for train and test sets
Step 14: Add new $\Theta\tau$ days of data as X'_i
Step 15: Y'_i = Model.predict (X'_i)
Output: Trained LSTM model and predictions.

3.4. The proposed DRHP approach

In this approach, the planning scheme is divided into multiple horizons. Each horizon consists of Θ operational periods and Ξ forecasting periods, with each period spanning τ days. Fig. 2 presents an RH planning framework with one operational period and Ξ forecasting periods. The developed MILP model optimizes production planning over each horizon. The resulting plan is mandatory to be implemented within the operational periods, yet it can be modified for the forecasting periods. Upon finishing one set of operational periods, a new horizon is defined, and the optimization step is repeated.

Once a horizon is rolled (i.e., when its operational set ends), the production plan is updated based on the newly revealed market demand and recycling rate, as well as the system's current status including the inventory level and the quantity of received raw materials (see Fig. 3).

Algorithm 2. Is the detailed representation of the developed DRHP approach, where h_o and h_f denote the operational set and the forecasting set of the horizon h, respectively.

Algorithm 2. ***Proposed DRHP Approach.***

Input: RH-based model and its parameters, trained rolling LSTM model, τ, Θ, and Ξ.
Step 1: Determine $H = \left\lfloor \frac{T}{\Theta\Xi} \right\rfloor - 1$
Step 2: ***For*** $p \in P$
  Train the LSTM model on both historical demand and recycling rate based on $\Theta\tau$
end for
Step 3: Let $h = 1$
Step 4: ***For*** $p \in P$
  Update $D_{pt}^{h_o} \forall t \in t_{h_o}$
  Fine tune the LSTM model (if needed) and forecast $RR_p^{h_f}$
  Set recycling rate on $RR_p^{h_f}$
  for $t \in t_{h_o}$
    Set demand on $D_{pt}^{h_o}$
  end for
  for $t \in t_{h_f}$
    Fine tune the LSTM model (if needed) and forecast $D_{pt}^{h_f}$
    Set demand on $D_{pt}^{h_f}$
  end for
end for
Step 5: Solve the RH-based model for horizon h
Step 6: Return decision variables and production cost related to horizon h
Step 7: Let $h = h + 1$
Step 8: ***While*** $h \le H$
  Update $RR_p^{(h-1)_f}$
  Calculate L_p^h using the actual $RR_p^{(h-1)_f}$
  Calculate $Invr_p^h, Invf_p^h$, and X_p^h using $invr_p^{(h-1)_o}$, $invf_p^{(h-1)_o}$, and $x_p^{(h-1)_o}$
  Go back to Step 4
end while
Output: Developed production plan and corresponding ATC.

3.5. Robust optimization

To evaluate the performance of the proposed DRHP approach, its static MILP counterpart model is also developed (refer to Appendix A for details). To tackle the uncertainty of demand and recycling rate in the static approach, robust optimization is adopted.

Robust optimization is particularly beneficial when accurately estimating the probability distribution of uncertain parameters is challenging or when decision-makers adopt a risk-averse strategy (Ben-Tal and Nemirovski, 1999; Bertsimas et al., 2004; Kisomi et al., 2016; Mohseni and Pishvaee, 2020; Tosarkani and Amin, 2020; Goerigk and Kurtz, 2023). In this approach, only the maximum and minimum possible values of uncertain parameters are required to model uncertainty. Simply put, variation in a parameter is represented as an uncertainty set that comprises a range of possible scenarios without explicitly defining a probability distribution. The objective is to offer a solution that performs well across all scenarios in the uncertainty set, ensuring robustness against fluctuations in uncertain parameters. The primary advantage of an uncertainty-based robust optimization approach is its ability to create a solution that is less sensitive to specific realizations of uncertain parameters within their uncertainty sets. This provides a degree of feasibility assurance when confronting unforeseen situations.

Over the last years, several uncertainty sets (e.g., box uncertainty set, ellipsoidal uncertainty set, polyhedral uncertainty set, box and ellipsoidal uncertainty set, box and polyhedral uncertainty set, ellipsoidal and polyhedral uncertainty set) and their robust counterpart formulation are introduced and discussed in the literature (Ben-Tal and Nemirovski, 1999; Bertsimas et al., 2004; Mohseni and Pishvaee, 2020).

3.5.1. Box uncertainty set-based robust optimization

To address the uncertainty in the static counterpart model, the box uncertainty set-base robust optimization is employed. To illustrate how this approach works, consider the following linear optimization model:

$$Min\left\{\sum_j \widetilde{c}_j x_j \middle| \sum_j \widetilde{a}_{ij} x_j - \widetilde{b}_i \le 0;\ \forall i\right\} \tag{22}$$

In this model, $\widetilde{c}_i$, $\widetilde{a}_{ij}$ and $\widetilde{b}_i$ represent uncertain parameters as follows:

$$\widetilde{c}_j = c_j + \rho_j \widehat{c}_j;\ \forall j \in J_i \tag{23}$$

$$\widetilde{a}_{ij} = a_{ij} + \rho'_{ij} \widehat{a}_{ij};\ \forall j \in J_i \tag{24}$$

$$\widetilde{b}_i = b_i + \rho''_i \widehat{b}_i \tag{25}$$

Where c_i, a_{ij}, and b_j denote the nominal values of the uncertain parameters, while $\widehat{c}_i$, $\widehat{a}_{ij}$, and $\widehat{b}_j$ represent the scales of uncertainty. The Robust Counterpart (RC) of the previous model is formulated as follows:

$$Min\left\{\sum_j c_j x_j + \left[\max_{\rho \in U}\left\{\sum_{J_i} \rho_j \widehat{c}_j x_j\right\}\right] \middle| -b_i + \sum_j a_{ij} x_j + \left[\max_{\rho \in U}\left\{-\rho''_i \widehat{b}_i + \sum_{J_i} \rho'_{ij} \widehat{a}_{ij} x_j\right\}\right] \le 0 \forall i\right\} \tag{26}$$

In this formulation, ρ_j, ρ'_{ij} and ρ''_i are random variables symmetrically distributed in the interval [-1,1]. To ensure the model feasibility, an uncertainty set U is defined for any ρ. A robust solution is characterized by its ability to remain feasible across all potential scenarios in the uncertainty set. This study adopts the box uncertainty approach as described by Li et al. (2011), where the uncertainty set is defined based on the ∞-norm of the random variable ρ, incorporating the adjustable parameter Ψ to control the size of the uncertainty set.

$$U_{box} = \{\rho | \|\rho\|_\infty \le \Psi\} = \{\rho | |\rho_j| \le \Psi;\ \forall j \in J_i\} \tag{27}$$

Using the box uncertainty set, the RC is reformulated as the following expression:

$$Min\left\{\sum_j c_j x_j + \Psi\left[\sum_{J_i} (\widehat{c}_j |x_j)\right] \middle| -b_i + \sum_j a_{ij} x_j + \Psi\left[\sum_{j \in J_i} (\widehat{a}_{ij} |x_j) + \widehat{b}_i\right] \le 0;\ \forall i\right\} \tag{28}$$

The adoption of the box uncertainty set ensures maximum protection against uncertainty, albeit at the expense of high costs for robustness in practical applications. This uncertainty set imposes stricter constraints compared to the original linear programming, as it enables all uncertain parameters to take their worst values at the same time. The conservatism of the box uncertainty set is evident in its capacity to account for the rare occurrence of the worst possible combination of uncertain parameters at the same time.

4. Results and sensitivity analysis

In this section, the performance of the proposed DRHP approach is evaluated across a set of instances in terms of different criteria such as inventories, backlog, late-order, and ATC.

4.1. Data and parameter setting

The scarcity of accurate real-world data has been a daunting challenge in many scientific studies. Although efforts have been made to provide public datasets, the availability of reliable data is constrained by factors such as data accessibility, scale, and privacy (Zhang et al., 2018). To address this challenge, generating synthetic datasets is a promising solution, allowing researchers to develop innovative data-driven models and evaluate them. Synthetic data features are categorized into two distinct statistical components: level and pattern. The level component encompasses key statistical attributes such as mean, scale, and variance, while the pattern component captures the underlying trend in the data (Zhang et al., 2018).

A survey conducted by Jones et al. (2007) investigated drinking water consumption in British Columbia, Canada, revealing an average daily intake of 1.0 L per capita. Notably, 23 % of respondents relied on bottled water as their primary drinking water source, accounting for at least 75 % of their daily intake. With the reported population of British Columbia at 5,437,722 as of April 1, 2023, it can be perceived that 1,250,676 individuals rely on bottled water as the main source of daily intake. Having that said, at least 938,007 L of daily water intake are supplied through bottled water. Assuming each bottle holds 0.5 L, the average daily demand for bottled water totals around 1,900,000 units. With an assumption of a market share of 1.5 %, the average demand for a specific manufacturing plant is 28,500 bottles per day. To synthesize historical data, the pattern in daily demand is generated similar to studies in the literature (Wong et al., 2010; Farah et al., 2019; Du et al., 2020; Nasser et al., 2020). The details of the data generation process are provided at https://github.com/RaziehLarizadeh/DRHP under the DS section, which is determined based on average demand, seasonality pattern and noise. Fig. 4.a) illustrates three synthetic scenarios for daily bottle demand (i.e., DS1, DS2, and DS3) over 365 days with the same average but different variations and patterns.

Antonopoulos et al. (2021) analyzed recycling activities in a number of different facilities and reported sorting rate ranging from 45 % to 97 % and recycling rate from 63 % to 95 %, which vary depending on the efficiency of the recycling process. Fig. 4.b) demonstrates three synthetic scenarios for daily recycling rate (i.e., RRS1, RRS2, and RRS3) over one year, with trends imitated based on studies by Ahmed et al. (2022) and Williams and Phillips (2022). The details of the data generation process are provided at https://github.com/RaziehLarizadeh/DRHP under the RRS section, where a random recycling stream with specific characteristics is selected each time to supply the recycled raw materials.

With the assumption of access to the first 180 days of historical data on demand and recycling rate, the goal is to develop a production plan for the upcoming 185 days, denoted as the Horizon era. Appendix B (Table B1) represents the parameter setting for the RH-based MILP model. Henceforth, it is assumed that $\Theta = \Xi = P = 1$ indicating rolling duration is equal to $\Theta\tau = 1 \times \tau = \tau$. Different scenarios of τ, i.e., {2, 4, 6, …, 22}, are investigated to evaluate the impact of rolling duration on forecasting accuracy and ATC.

To evaluate the role of the LSTM model in managing uncertainty in the proposed DRHP approach, its impact is benchmarked against two alternatives. The first alternative involves eliminating the uncertainty assumption and using the exact actual values of uncertain parameters. The second one relies on the average values of uncertain parameters (derived from historical data) as predictions of their future values.

4.2. Forecasting accuracy measures

Various measures can be used to evaluate the performance of time-series forecasting models, each assessing the outputs in terms of specific criteria. In this study, the performance of the LSTM model is analyzed using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) computed in Equations (29) to (31), respectively. Y_i and $\widehat{Y}_i$ are the actual and forecasted values of the i th observation, respectively and the number of total observations is denoted by n.

$$MSE = \frac{\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2}{n} \tag{29}$$

$$RMSE = \sqrt{MSE} \tag{30}$$

$$MAE = \frac{\sum_{i=1}^{n}|Y_i - \widehat{Y}_i|}{n} \tag{31}$$

MSE determines the average of the squared predicted errors. RMSE is the standard deviation of the prediction errors, and it is scale-dependent and sensitive to outliers. Finally, MAE is the absolute value of the difference between the actual value and the forecasted value. MAE estimates expected error from the forecast on average, and it is beneficial especially when there are outliers in the data.

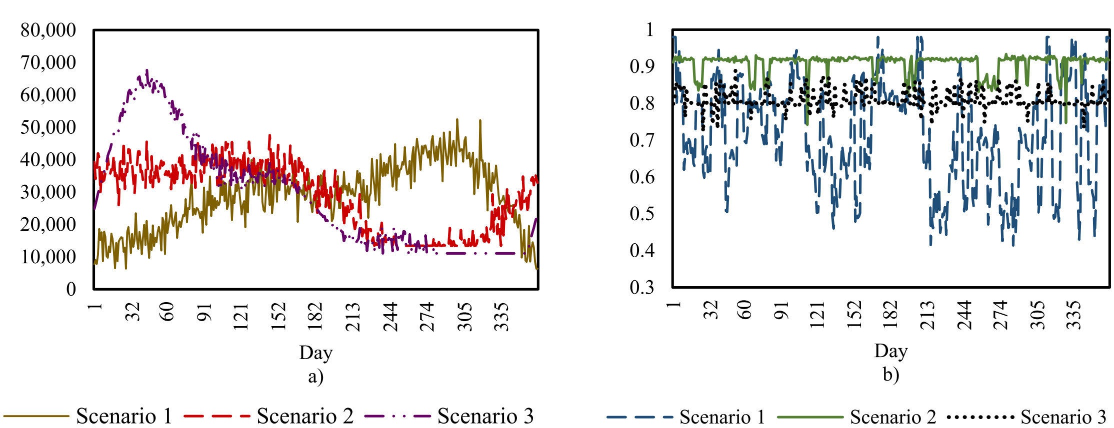


Fig. 4. The synthetic scenarios of a) market demand, and b) recycling rate.

Table 2
Performance of the rolling LSTM model in terms of MSE, RMSE, and MAE for predicting recycling rate and market demand across different scenarios of rolling duration.

		MSE-Train			**MSE**-Test			**MSE**-Train			**MSE**-Test		
		RRS1	RRS2	RRS3	RRS1	RRS2	RRS3	DS1	DS2	DS3	D-S1	DS2	DS3
Rolling duration	2	0.0353	0.0201	0.0311	0.0345	0.0117	0.0240	0.0127	0.0171	0.0040	0.0113	0.0204	0.0041
	6	0.0439	0.0229	0.0287	0.0468	0.0397	0.0274	0.0098	0.0143	0.0067	0.0112	0.0161	0.0062
	10	0.0474	0.0235	0.0290	0.0485	0.0250	0.0253	0.0102	0.0143	0.0112	0.0099	0.0145	0.0102
	14	0.0481	0.0258	0.0285	0.0534	0.0289	0.0319	0.0097	0.0141	0.0231	0.0114	0.0145	0.0198
	18	0.0473	0.0248	0.0305	0.0566	0.0198	0.0328	0.0088	0.0140	0.0229	0.0101	0.0143	0.0175
	22	0.0572	0.0202	0.0284	0.0696↓	0.0176↓	0.0354↓	0.0090	0.0136	0.0202	0.0104	0.0142	0.0268↓
		RMSE-Train			**RMSE**-Test			**RMSE**-Train			**RMSE**-Test		
		RRS1	RRS2	RRS3	RRS1	RRS2	RRS3	DS1	DS2	DS3	DS1	DS2	DS3
Rolling duration	2	0.1881	0.1418	0.1766	0.1857	0.1084	0.1549	0.1127	0.1310	0.0633	0.1065	0.1431	0.0641
	6	0.2096	0.1516	0.1696	0.2163	0.1994	0.1657	0.0992	0.1196	0.0821	0.1061	0.1269	0.0792
	10	0.2177	0.1534	0.1705	0.2202	0.1581	0.1592	0.1011	0.1192	0.1062	0.0998	0.1206	0.1010
	14	0.2194	0.1607	0.1690	0.2312	0.1702	0.1787	0.0989	0.1187	0.1522	0.1068	0.1208	0.1409
	18	0.2175	0.1575	0.1748	0.2379	0.1409	0.1812	0.0942	0.1183	0.1515	0.1009	0.1197	0.1325
	22	0.2392	0.1421	0.1685	0.2639↓	0.1329↓	0.1883↓	0.0948	0.1167	0.1424	0.1020	0.1194	0.1639↓
		MAE-Train			**MAE**-Test			**MAE**-Train			**MAE**-Test		
		RRS1	RRS2	RRS3	RRS1	RRS2	RRS3	DS1	DS2	DS3	D-S1	DS2	DS3
Rolling duration	2	0.1484	0.0790	0.1259	0.1372	0.0681	0.0983	0.0888	0.1056	0.0519	0.0864	0.1173	0.0551
	6	0.1780	0.0925	0.1216	0.1934	0.1134	0.1207	0.0798	0.0975	0.0644	0.0839	0.1063	0.0622
	10	0.1877	0.1013	0.1272	0.1918	0.1045	0.1198	0.0792	0.0984	0.0777	0.0777	0.0995	0.0744
	14	0.1887	0.1150	0.1242	0.1998	0.1226	0.1357	0.0777	0.0979	0.1044	0.0836	0.1000	0.0940
	18	0.1892	0.1064	0.1316	0.2061	0.0976	0.1416	0.0745	0.0975	0.1014	0.0820	0.0985	0.0840
	22	0.1961	0.0864	0.1167	0.2239	0.0708	0.1327	0.0753	0.0969	0.0971	0.0834	0.0983	0.1078

↓ shows an overall decline in prediction accuracy by increasing the rolling duration parameter.

4.3. Performance evaluation

In this section, the performance of the proposed DRHP approach is explored across multiple instances characterized by different scenarios of demand, recycling rate, and rolling duration.

4.3.1. Evaluation of the LSTM model

The rolling LSTM model is trained separately on the first 180 days, considered as historical data, of different scenarios of demand and recycling rate. The general setting of the LSTM model is summarized in Appendix B (Table B2). All the experiments are performed on a personal computer with an Intel Core i5 (2.4 GHz) CPU, 16 GB memory, and Windows 10 operating system. Table 2 demonstrates the performance of Algorithm 1 in terms of the MSE, RMSE, and MAE values for various scenarios of market demand and recycling rate under different rolling durations. In this study, the rolling duration is determined based on the raw material preparation efficiency (i.e., the maximum time required to source raw materials, whether from virgin plastics or recycled plastics), and directly affects the system's updating strategy. For normalized data, a desired range of error-based metrics should be near zero, indicating close alignment between the model's predictions and actual values. Having said this, the results reported in Table 2 demonstrate the satisfactory performance of the trained rolling LSTM model on both demand and recycling rate datasets. However, it can be seen that forecasting accuracy in RRS1, RRS2, RRS3, and DS3 declines as the rolling duration increases. To determine the significance of rolling duration on the output of the DRHP approach, this parameter is examined regarding its effects on ATC, backlog, late orders, and inventory levels over the subsequent 185 days of data in the next sections.

4.3.2. Evaluation of the DRHP approach

In this section, the performance of the DRHP approach is benchmarked against its robust static counterpart (detailed in Appendix A) in terms of ATC. This comparison is performed across nine different instances characterized by various combinations of demand and recycling rate scenarios (i.e., DS1-RRS1, DS1-RRS2, …, DS3-RRS3). The worst-case scenario of the robust static counterpart in terms of the expected total production cost (Appendix A, Eq. 32) occurs when future demand and future recycling rate are set at the maximum and minimum of their historical data, respectively (denoted by $D' = D'\text{max}$ and $R' = R'\text{min}$). However, determining the worst-case scenario in terms of ATC (Appendix A, Eq. 49) is not straightforward. The reason is that ATC depends not only on historical data but also on actual future values of demand and recycling rate. In this regard, all four potential worst-case scenarios from the ATC standpoint, considering demand and recycling rate at their historical extremes (i.e., (i) $D'\text{min}-R'\text{min}$, (ii) $D'\text{min}-R'\text{max}$, (iii) $D'\text{max}-R'\text{min}$, and (iv) $D'\text{max}-R'\text{min}$,), are taken into account in this study.

The goal is to demonstrate a meaningful difference in performance between the DRHP approach and its robust static counterpart from the ATC standpoint. The significance of this hypothesis is examined using the Friedman Test (Zamri et al., 2022; Zamri et al., 2024). Detailed findings from this non-parametric statistical test are presented in Table 3. The null hypothesis (i.e., H_0) is that there is no significant

Table 3
Friedman test results in terms of ATC with a confidence interval of 0.95.

	DRHP	Robust static ($D'min - RR'\text{min}$)	Robust static ($D'min - RR'\text{max}$)	Robust static ($D'max - RR'\text{min}$)	Robust static ($D'max - RR'\text{max}$)
H_0	no significance between the DRHP approach with other comparative models				
df	4				
χ^2	121.8182				
p_value	2.1841×10^{-25}				
Conclusion	Reject H_0				

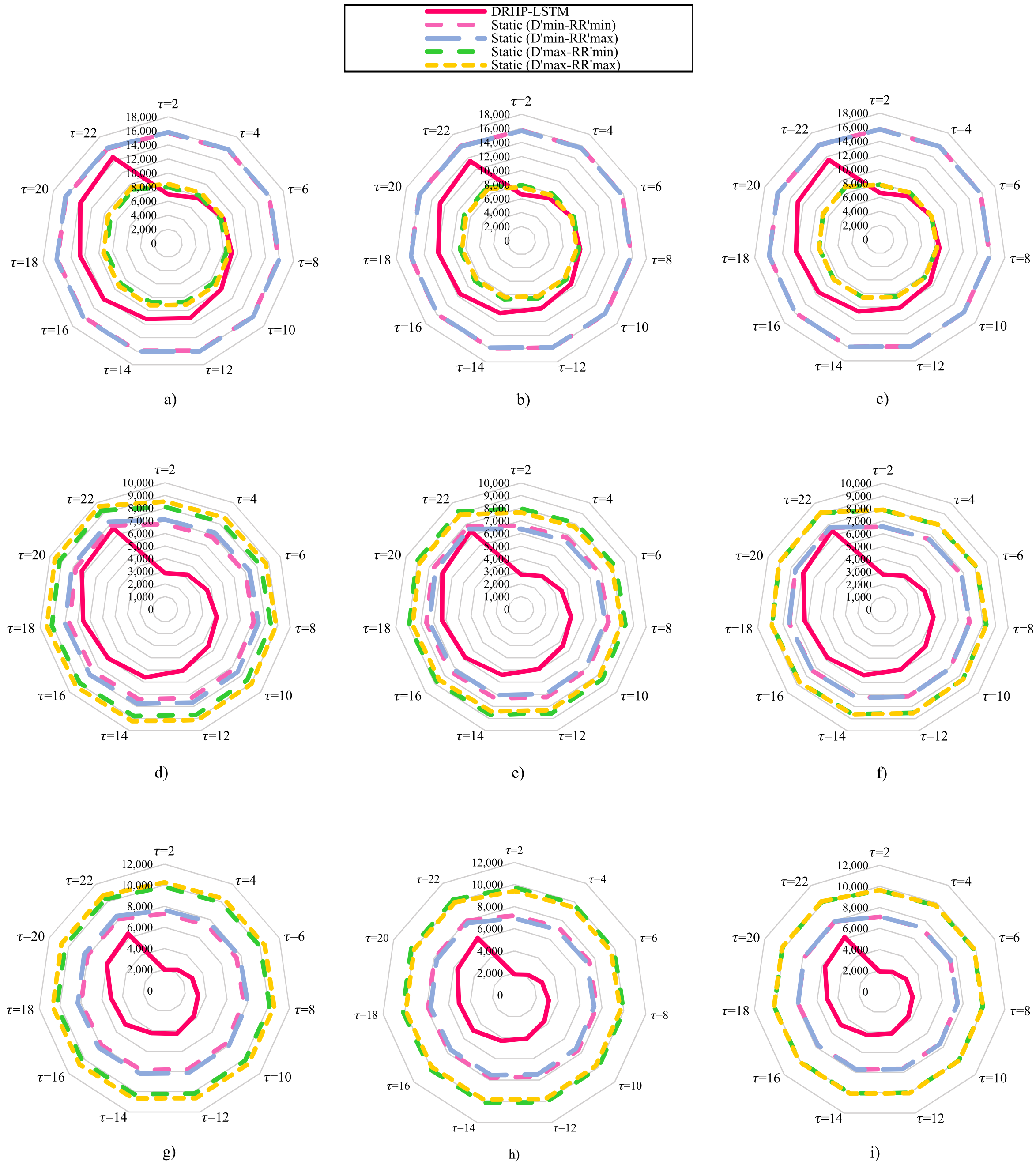


Fig. 5. Comparison between the DRHP approach and the robust static models in terms of ATC (thousands) across different scenarios: a) DS1-RRS1, b) DS1-RRS2, c) DS1-RRS3, d) DS2-RRS1, e) DS2-RRS2, f) DS2-RRS3, g) DS3-RRS1, h) DS3-RRS2, and i) DS3-RRS3.

difference in performance between the DRHP approach and other comparative models. The decision is made at a significance level of $\alpha = 0.05$. Note that χ^2 is the test statistics following the Chi-Square distribution with the degree of freedom (k-1), $df = 4$. Given that the *p*-value is much less than α, H_0 is rejected. This indicates that, in general, the different approaches perform significantly differently from each other or in other words, they do not perform equally.

To verify that the DRHP approach is the superior one, Fig. 5 provides a detailed comparison between the DRHP approach and the robust static models (with different assumptions of worst-case scenarios) in terms of ATC. The results show the outperformance of the DRHP approach over the robust static models, particularly in the instances characterized by DS2 and DS3, as evidenced by a significant reduction in ATC. However, when dealing with DS1, the comparison between the approaches becomes nuanced, depending on the rolling duration. Moreover, the results highlight that ATC remains relatively unaffected by increasing rolling duration in the robust static models. On the contrary, in the DRHP approach, ATC significantly differs across various settings of this parameter, indicating that smaller values of rolling duration result in better performance of the DRHP approach. This implies the significance of efficient raw material sourcing in the DRHP approach as the rolling duration parameter and, consequently, updating frequency are determined based on the efficiency of raw material delivery/preparation. Table 4 illustrates the percentage reduction in ATC by employing the DRHP approach instead of the robust static models in which the worst-case scenario is defined by $D'\text{min}-R'\text{max}$ and $D'\text{max}-R'\text{min}$. The results indicate that the DRHP approach surpasses its robust static counterpart by reducing ATC up to nearly 65 % and 80 % in the instances dealing

Table 4
Reduction in ATC (percentage) by adopting the DRHP approach instead of the robust static approach. The numbers in bold highlight the greatest reductions.

Instance characterized by	Worst-case scenario	Raw material preparation efficiency/rolling duration										
		2	4	6	8	10	12	14	16	18	20	22
DS1-RRS1	$D'min - RR'max$	56.18	51.49	45.89	42.77	37.47	30.59	30.09	24.14	21.14	14.42	9.57
	$D'max - RR'min$	13.45	5.30	−4.05	−8.53	−16.88	−28.11	−27.70	−37.40	−41.87	−51.99	−58.90
DS1-RRS2	$D'min - RR'max$	57.97	53.81	49.39	46.14	40.61	36.52	32.14	27.09	24.66	20.06	15.76
	$D'max - RR'min$	16.75	9.62	2.47	−2.34	−11.19	−17.35	−24.14	−32.23	−35.73	−42.14	−48.18
DS1-RRS3	$D'min - RR'max$	57.65	53.49	48.90	45.68	40.33	36.06	32.81	27.71	24.62	19.54	15.77
	$D'max - RR'min$	14.83	7.63	0.08	−4.69	−13.29	−19.79	−24.54	−32.82	−37.52	−44.84	−49.95
DS2-RRS1	$D'min - RR'max$	59.83	54.74	50.17	44.61	40.47	34.24	27.95	24.73	18.14	10.67	7.73
	$D'max - RR'min$	**64.76**	60.19	56.08	51.12	47.42	41.81	36.21	33.39	27.66	21.11	18.19
DS2-RRS2	$D'min - RR'max$	56.89	51.93	46.29	40.54	35.85	29.09	23.75	18.98	13.83	8.16	2.89
	$D'max - RR'min$	**65.65**	61.532	56.83	52.04	48.13	42.43	37.86	33.81	29.51	24.77	20.00
DS2-RRS3	$D'min - RR'max$	57.81	52.58	47.18	41.72	36.93	30.96	25.80	21.32	16.64	9.29	3.92
	$D'max - RR'min$	**64.84**	60.37	55.73	51.04	46.93	41.76	37.24	33.36	29.37	23.13	18.20
DS3-RRS1	$D'min - RR'max$	74.18	69.92	63.40	59.43	54.05	47.69	48.75	40.73	38.32	28.42	24.07
	$D'max - RR'min$	**79.94**	76.47	71.22	67.92	63.49	58.23	58.95	52.37	50.27	42.07	38.32
DS3-RRS2	$D'min - RR'max$	73.04	68.73	61.55	56.83	51.391	45.25	42.82	35.25	33.74	26.86	20.99
	$D'max - RR'min$	**80.77**	77.52	72.15	68.52	64.32	59.57	57.53	51.65	50.26	44.81	40.11
DS3-RRS3	$D'min - RR'max$	73.33	69.142	62.13	57.56	52.31	46.12	44.44	37.59	36.00	27.92	22.75
	$D'max - RR'min$	**80.23**	76.97	71.56	67.93	63.76	58.84	57.35	51.86	50.40	43.89	39.63

Table 5
Amount of raw material (kg) sourced via the virgin plastic stream in the robust static approach.

Instance characterized by	Worst-case scenario	Raw material preparation efficiency/rolling duration					
		2	6	10	14	18	22
DS1	$D'min - RR'min$	116,168	113,629	111,090	108,550	106,011	103,472
	$D'max - RR'min$	549,000	537,000	525,000	513,000	501,000	499,000
DS2	$D'min - RR'min$	475,049	464,666	454,282	443,898	433,515	423,131
	$D'max - RR'min$	549,000	537,000	525,000	513,000	501,000	499,000
DS3	$D'min - RR'min$	454,407	444,474	434,542	424,610	414,677	404,745
	$D'max - RR'min$	549,000	537,000	525,000	513,000	501,000	499,000

with DS2 and DS3, respectively. In the cases characterized by DS1, the DRHP approach loses superiority for rolling durations exceeding four days. This happens only when the worst-case scenario in the robust static approach is defined by setting future demand at the maximum value of the historical data.

To comprehensively evaluate the performance of the two approaches, a sustainability performance assessment is also conducted across the nine instances. The results highlight a significant difference in the raw material sourcing strategies between the two approaches. The DRHP approach demonstrates a commitment to sustainability by opting for recycled plastics, whereas the robust static counterpart mostly relies on virgin plastics. Table 5 presents the amount of raw material sourced from virgin plastics in the robust static approach under two different worst-case scenarios. Findings imply that the DRHP approach is a more environmentally friendly approach when compared with its robust static counterpart.

A key feature of the DRHP approach is the integration of the forecasting model with the RH-based planning model, allowing accurate and continuously updated predictions of demand and recycling rate to be fed into the planning process. Fig. 6 illustrates the importance of using an accurate forecasting model to manage uncertainty in the DRHP approach. This assessment benchmarks the significance of the LSTM model against two alternatives: (i) assuming the future values of uncertain parameters are equal to the historical average and (ii) having perfect knowledge of the exact future values. Results show that the proposed DRHP approach closely resembles the performance of the RH-based model with perfect knowledge of future values. Also, the DRHP approach notably outperforms the RH-based model that relies on the average of historical data. This emphasizes the importance of coupling the RH-based model with an accurate forecasting model to manage uncertainty challenges. Moreover, it is evident that the loss of forecasting accuracy (denoted by LSTM to Average) results in more substantial cost increases as the rolling duration grows. This observation implies that when it is not feasible to set smaller values for the rolling duration, adopting the proposed DRHP approach becomes even more critical. Finally, it can be seen that variations in recycling rate scenarios have minimal impact on ATC, with changes being more influenced by demand scenarios.

When addressing optimization problems, accurately identifying the primary cost drivers is essential for informed decision-making. This allows planners to diagnose critical inefficiencies within the system and formulate targeted solutions to address these vulnerabilities. Fig. 7 depicts the performance of the DRHP approach in terms of backlog, raw material inventory, final product inventory, and late-order. By leveraging the LSTM model, the RH-based model results in inventory and late-order levels similar to those obtained assuming the exact future values of uncertain data. However, backlog remains consistent across different strategies of uncertainty handling. This observation suggests that in cases where production plants struggle with inefficiencies related to inventory or late-order management, the proposed DRHP approach

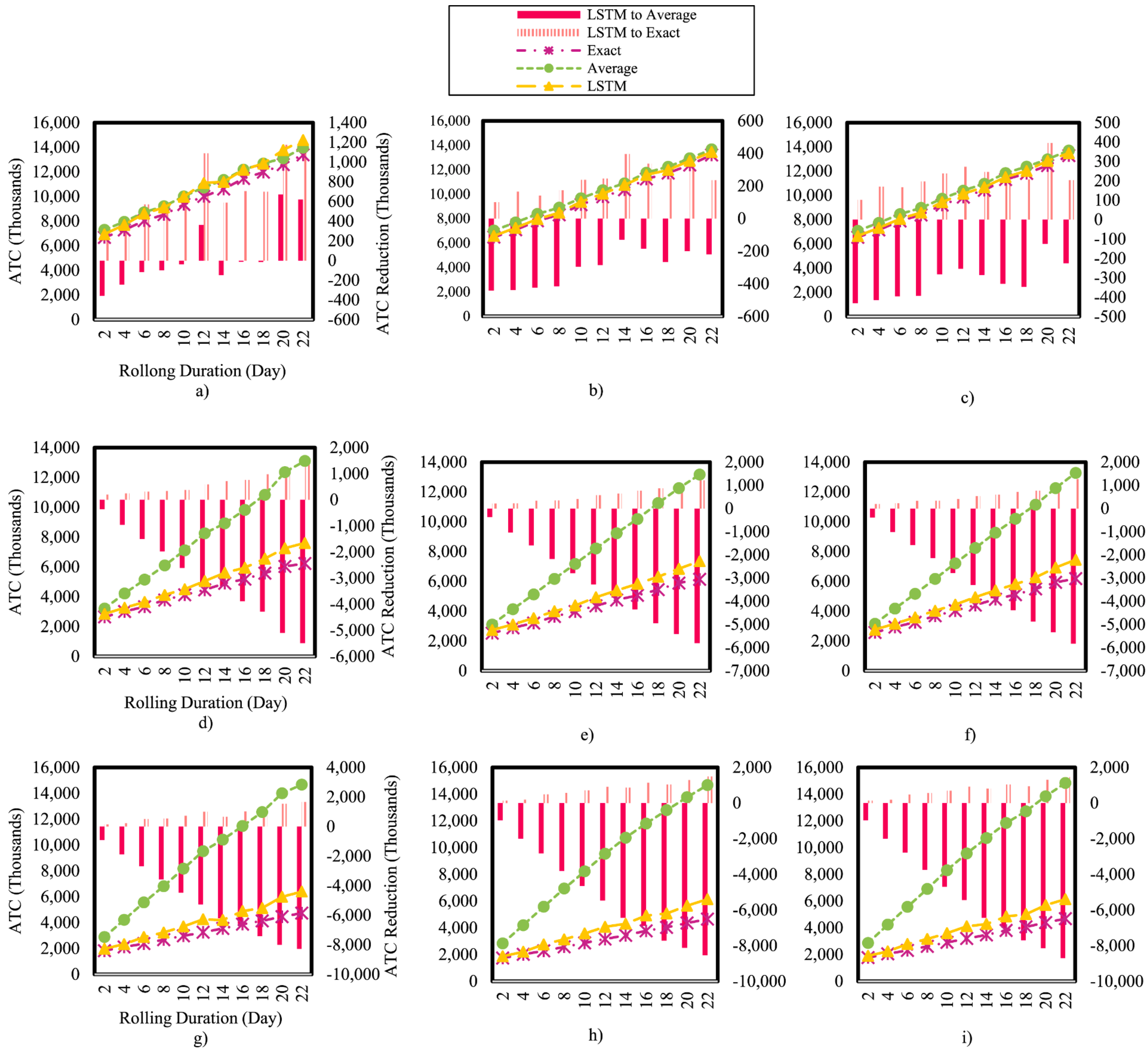


Fig. 6. Performance of the DRHP approach in terms of ATC across different instances characterized by: a) DS1-RRS1, b) DS1-RRS2, c) DS1-RRS3, d) DS2-RRS1, e) DS2-RRS2, f) DS2-RRS3, g) DS3-RRS1, h) DS3-RRS2, and i) DS3-RRS3.

holds promise. Note that the data presented in Fig. 7 is based on the average results of the different nine instances explained earlier.

4.4. Sensitivity analysis

Sensitivity analysis aims to evaluate how changes in input parameters or assumptions of a problem can affect the outputs. This offers decision-makers insight into the robustness and reliability of their approach. In this regard, an analysis is conducted in this section to explore the sensitivity of the DRHP approach to the rolling duration parameter.

Fig. 8 illustrates the impact of various rolling durations on backlog, raw material inventory, final product inventory, and late-order. The results show that an increase in rolling duration has the most significant negative effect on raw material inventory, followed by late orders and final product inventory, while having no substantial impact on the backlog level. This indicates that the rise in ATC for longer rolling durations (as shown in Fig. 6) is primarily driven by increased inventory and late-order levels, rather than backlog. In summary, shorter rolling durations, or in other words, more frequent system updates, are highly advantageous in minimizing costs associated with late orders and inventory levels, thereby reducing ATC.

The developed RH-based model comprises multiple cost parameters related to raw material sourcing, production, recycling, backlog, late-order, and inventory management. Sensitivity analysis on the cost-related parameters provides insights into the robustness of the proposed DRHP approach, enabling the identification of critical components and their potential risks. In this regard, an analysis is conducted to examine the impact of various values of inventory costs ($Cinvr^h_{pt}$, $Cinvf^h_{pt}$), penalty costs (Cbl^h_{pt}, Cdo^h_p), and raw material sourcing costs (Cvp^h_p, Cr^h_p) on ATC (see Fig. 9). The results indicate that penalty costs lead to an increase of production costs at higher values as the DRHP approach does not address delays. However, increasing inventory and raw material sourcing costs have only a marginal influence on ATC, implying that the proposed approach can adjust the plan to prevent a drastic increase in total production cost. This verifies the findings of the previous analyses where it is mentioned that the DRHP approach holds promise in addressing challenges related to inefficiencies in inventory or late-order management. Note that the data presented in Figs. 8 and 9 are based on the average results of the nine instances explained earlier.

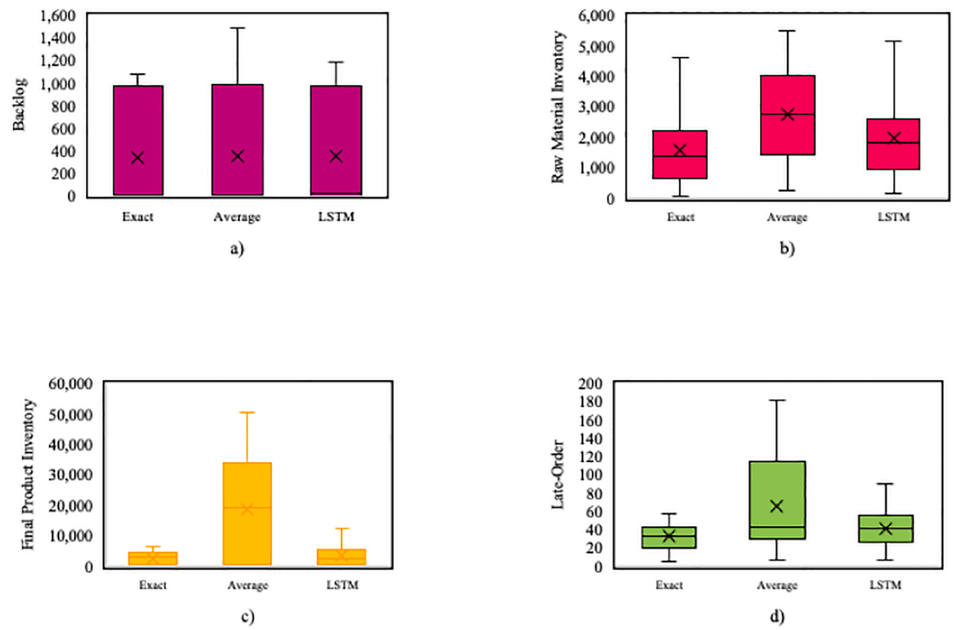


Fig. 7. Performance of the DRHP approach in terms of: a) backlog (thousands), b) raw material inventory (thousands), c) final product inventory (thousands), and d) late-order (thousands).

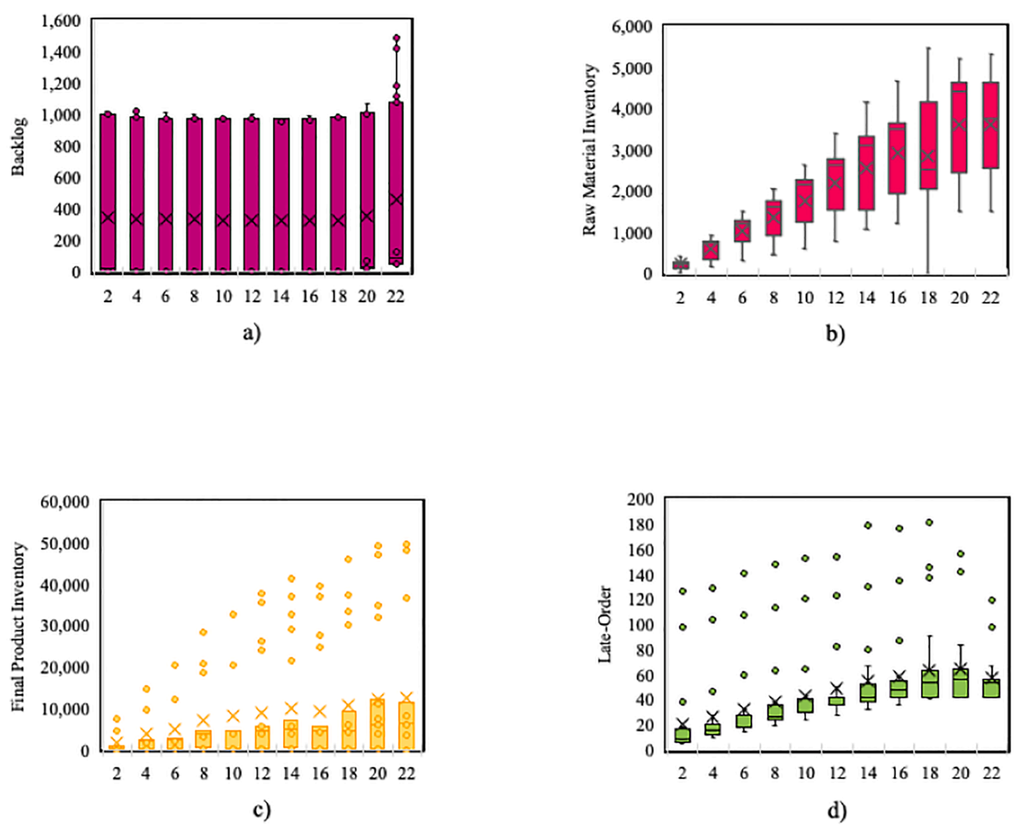


Fig. 8. Impact of rolling duration on a) backlog (thousands), b) raw material inventory (thousands), c) final product inventory (thousands), and d) late-order (thousands) in the DRHP approach.

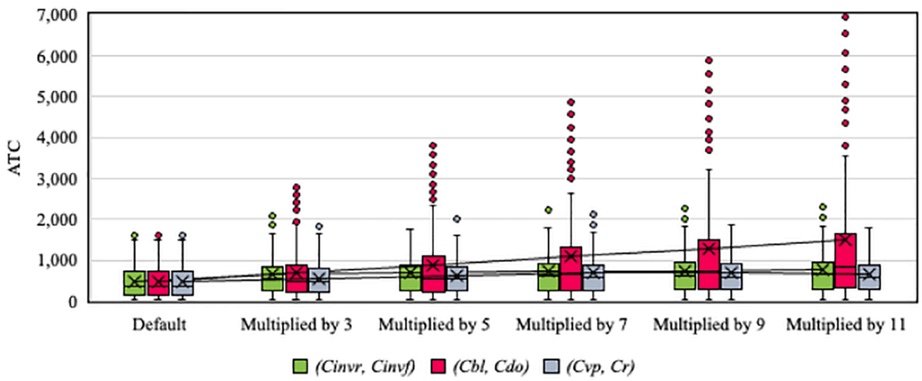


Fig. 9. Sensitivity analysis on different elements of production cost in terms of ATC (thousands).

5. Conclusions and future research

Developing an efficient production plan in the plastic industry requires embracing recycling practices and considering the dynamic nature of key parameters. To achieve this, the current study develops a dynamic DRHP approach for sustainable production planning aiming to incorporate recycling practices, fulfill market demand at the minimum possible production cost, and consider the uncertainty of market demand and recycling rate. This approach couples a novel RH-based mathematical model with a rolling LSTM model. The RH-based production planning is formulated as a multi-product, multi-period MILP model, incorporating various production-related components. This model assists decision-makers in deciding on raw material sourcing stream (i.e., virgin or recycled), production scheduling, inventory management, and backlog and late-order handling. The rolling LSTM model adjusts the predictions for future demands and recycling rates, consequently assisting in modifying the production plan as horizons progress. After each horizon roll, the entire system is re-assessed, and the production plan is updated based on the current status of the system, recently revealed data, and prediction of future trends in uncertain parameters. The accuracy of future forecasting is a determinant in this approach, impacting both current and future planning outcomes. To evaluate the performance of the proposed DRHP approach, it is benchmarked against a robust static MILP counterpart model to illustrate the impact of dynamic planning.

Findings demonstrate the superiority of the DRHP approach over the robust static counterpart from both economic and environmental points of view. Moreover, findings underscore the significance of the rolling duration parameter on the DRHP approach's performance, by affecting both the RH-based model and the LSTM model. Results illustrate that the total production cost substantially increases as the rolling duration extends. This implies that by improving the raw material preparation efficiency and consequently shortening the rolling duration, the performance of the DRHP approach significantly improves. Furthermore, findings indicate that an increase in total production cost under long rolling durations mainly stems from the backlog level. The proposed DRHP approach significantly mitigates inventory- and late-order-related costs, while the backlog does not change as significantly. Results show that improving the LSTM forecasting accuracy significantly improves the RH-based model's outcomes. This improvement is more substantial when dealing with a longer rolling duration. This highlights that applying a data-driven technique is crucial specifically when the rolling duration is set on longer durations. Aligning with previous results, sensitivity analysis indicates that penalty costs (i.e., Cbl_{pt}^{h}, Cdo_{p}^{h}) are the most influential cost components affecting ATC. However, the approach effectively confronts an increase in the costs related to inventory (i.e., $Cinvr_{pt}^{h}$, $Cinvf_{pt}^{h}$) and raw material sourcing (i.e., Cvp_{p}^{h}, Cr_{p}^{h}), showing the robustness of the approach when dealing with inventory and raw material sourcing challenges. It is worth mentioning that the proposed RH-based model can be extended to other industries seeking sustainable and dynamic production planning with only moderate adjustments to the industry-specific parameters and constraints.

The primary challenge in this research is the limited availability of perfect real-world data. Despite the efforts to generate synthetic datasets based on the available information and analyze various scenarios for more robust conclusions, access to 100 % real-world datasets is still needed for deeper insights. Studying historical real-world data may require developing advanced forecasting models to better capture the complexities and generate accurate predictions. Additionally, to develop a synchronized supply chain network, future research could integrate other elements of the network such as distribution centers and customers into the planning framework. The proposed DRHP approach could also be expanded into a multi-objective version to more strictly follow the sustainability goals by introducing an objective function particularly focused on minimizing CO2 emissions. Lastly, the DRHP approach holds potential for further extension into a more dynamic framework by incorporating deep reinforcement learning or control theory techniques to enhance real-time performance and adaptability.

CRediT authorship contribution statement

Razieh Larizadeh: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Babak Mohamadpour Tosarkani:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the editor and referees for the comments and feedback that improved the paper. This research was supported by the Discovery Grants provided by the Natural Sciences and Engineering Research Council of Canada (grant# RGPIN-2021-02912).

Appendix A

In the static counterpart approach, the entire production scheme is planned all at once in advance. Orders for raw materials, whether virgin plastics or plastic waste, can be placed at any time (without restrictions on the timing of orders). Similar to the DRHP approach, demand and recycling rate are uncertain parameters. The MILP model developed for the static counterpart is as follows:

Indices:

t	Days; $t = 1, \ldots, T$.
p	Type of raw materials/products; $p = 1, \cdots, P$.

Parameters:

υ	Raw material preparation efficiency, equivalent to rolling duration ($\Theta\tau$) in the DRHP approach (day).
D'_{pt}	Predicted demand of product p on day t (unit).
RR'_{pt}	Predicted recycling rate of plastic waste type p on day t (percentage).
$Pcap_{pt}$	Production capacity for product type p on day t (unit).
$Cinvr_{pt}$	Inventory cost of raw material type p on day t (\$ per kg).
$Cinvf_{pt}$	Inventory cost of final products type p on day t (\$ per unit).
$Cpro_{pt}$	Production cost of product type p on day t (\$ per unit).
Cbl_{pt}	Backlog cost of product type p on day t (\$ per unit).
Cvp_{pt}	Cost of virgin plastic type p on day t (\$ per kg).
Cwp_{pt}	Cost of plastic waste type p on day t (\$ per kg).
Cr_{pt}	Cost of recycling plastic waste type p on day t (\$ per kg).
Ccv_{pt}	Contract cost for virgin plastic type p on day t (\$).
Ccw_{pt}	Contract cost for plastic waste type p on day t (\$).
$Invr_p$	Inventory of raw material type p at the beginning of planning scheme (kg).
$Invf_p$	Inventory of final product type p at the beginning of planning scheme (unit).
α_p	Unitary raw material consumption for product type p (percent).
M	A very large positive number.
ε	A very small positive number.

Decision variables:

x_{pt}	Amount of virgin plastic type p ordered on day t (kg).
y_{pt}	Amount of plastic waste type p ordered on day t (kg).
z_{pt}	Amount of virgin plastic type p received on day t (kg).
l'_{pt}	Amount of raw material type p expected to be sourced through recycling stream on day t (kg).
pq_{pt}	Number of product type p produced on day t (unit).
sq_{pt}	Number of product type p sold on day t (unit).
$invr_{pt}$	Inventory of raw material type p on the end of day t (kg).
$invf_{pt}$	Inventory of final product type p on the end of day t (kg).
v_{pt}	Equals 1 if virgin material contributes to the raw material sourcing of product type p on day t, otherwise 0.
w_{pt}	Equals 1 if recycling contributes to the raw material sourcing of product type p on day t, otherwise 0.

Objective function:

$$MinZ_h = \sum_{p=1}^{P} \sum_{t=1}^{T} \left(Cvp_{pt}x_{pt} + Cwp_{pt}y_{pt} + Cr_{pt}l'_{pt} + Ccv_{pt}v_{pt} + Ccw_{pt}w_{pt} + Cinvr_{pt}invr_{pt} + Cinvf_{pt}invf_{pt} + Cpro_{pt}pq_{pt} + Cbl_{pt}\left(D'_{pt} - sq_{pt}\right) \right) \tag{A1}$$

Constraints:

$$pq_{pt} \le Pcap_{pt};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A2}$$

$$sq_{pt} \le D'_{pt};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A3}$$

$$z_{pt} = 0;\ \forall t \in \{1, \cdots, \upsilon\},\ \forall p \in \{1, \cdots, P\} \tag{A4}$$

$$z_{pt} = x_{p(t-\upsilon)};\ \forall t \in \{\upsilon + 1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A5}$$

$$l'_{pt} = 0;\ \forall t \in \{1, \cdots, \upsilon\},\ \forall p \in \{1, \cdots, P\} \tag{A6}$$

$$l'_{pt} = y_{p(t-\upsilon)}.RR'_{p(t-\upsilon)};\ \forall t \in \{\upsilon + 1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A7}$$

$$Invr_p - \alpha_p pq_{pt} = invr_{pt};\ t = 1,\ \forall p \in \{1, \cdots, P\} \tag{A8}$$

$$invr_{p(t-1)} - \alpha_p pq_{pt} + z_{pt} + l'_{pt} = invr_{pt};\ \forall t \in \{2, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A9}$$

$$Invf_p + pq_{pt} - sq_{pt} = invf_{pt};\ t = 1,\ \forall p \in \{1, \cdots, P\} \tag{A10}$$

$$invf_{p(t-1)} + pq_{pt} - sq_{pt} = invf_{pt};\ \forall t \in \{2, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A11}$$

$$\varepsilon.v_{pt} \le x_{pt} \le M.v_{pt};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A12}$$

$$\varepsilon.w_{pt} \le y_{pt} \le M.w_{pt};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A13}$$

$$x_{pt}, y_{pt}, z_{pt}, l_{pt}, pq_{pt}, sq_{pt}, invr_{pt}, invf_{pt}, v_{pt}, w_{pt} \in R^{+};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A14}$$

$$v_{pt}, w_{pt} \in \{0, 1\};\ \forall t \in \{1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A15}$$

Objective (A1) minimizes the production cost over the entire planning scheme. Constraint (A2) accounts for production capacity limits. Constraint (A3) ensures demand satisfaction. Constraints (A4) and (A5) specify the daily amount of raw material received through the virgin plastic stream. Constraints (A6) and (A7) determine the daily amount of raw material received through the recycling stream. Constraints (A8) and (A9) monitor the flow of raw material inventory throughout the entire planning process. Constraints (A10) and (A11) ensure a balanced inventory flow for final products. Constraints (A12) and (A13) determine the quantity of raw material sourced through virgin plastics and the recycling stream, respectively. Constraints (A14) and (A15) specify the range of the decision variables.

It is worth mentioning that the ATC in the static approach is also determined at the end of the planning scheme (after the disclosure of actual data of demand and recycling rate) as follows:

$$\begin{aligned} ATC :& \\ &= \sum_{p=1}^{P} \sum_{t=1}^{T} \Big(Cvp_{pt}x_{pt} + Cwp_{pt}y_{pt} + Cr_{pt}L_{pt} + Ccv_{pt}v_{pt} + Ccw_{pt}w_{pt} + Cinvr_{pt}invr_{pt} + Cinvf_{pt}invf_{pt} + Cpr_{pt}pq_{pt} + Cdo_{pt}Pdo_{pt}\left(l'_{pt} - L_{pt}\right) + Cbl_{pt}Ybl_{pt}\Big(D_{pt} \\ &\quad - sq_{pt}\Big) + Cex_{pt}\left(1 - Ybl_{pt}\right)_{pt}\left(sq_{pt} - D_{pt}\right)\Big) \end{aligned} \tag{A16}$$

Where

$$L_{pt} = 0;\ \forall t \in \{1, \cdots, \upsilon\},\ \forall p \in \{1, \cdots, P\} \tag{A17}$$

$$L_{pt} = y_{p(t-\upsilon)}.RR_{p(t-\upsilon)};\ \forall t \in \{\upsilon + 1, \cdots, T\},\ \forall p \in \{1, \cdots, P\} \tag{A18}$$

And.

L_{pt}	Actual Amount of raw material type p sourced through recycling stream on day t (kg).
D_{pt}	Actual demand of product p on day t (unit).
RR_{pt}	Actual recycling rate of plastic waste type p on day t (percentage).
Cdo_{pt}	Late-order cost for virgin plastic type p on day t ($ per kilogram).
Cex_{pt}	Cost of excessive production of product type p on day t ($ per unit).
Pdo_{pt}	Equals 1 if $L_{pt} \le l'_{pt}$, otherwise 0.
Ybl_{pt}	Equals 1 if $sq_{pt} \le D_{pt}$, otherwise 0.

Appendix B

Table B1
Parameter setting for the RH-based MILP model.

Parameter	Setting	Description
$Invr_p^h(h = 0)$	0	Initial raw material inventory (kg)
$Invf_p^h(h = 0)$	0	Initial final product inventory (unit)
$X_p^h(h = 0)$	0	Initial virgin raw material supply (kg)
$L_p^h(h = 0)$	0	Initial recycled raw material supply (kg)
$Pcap_{pt}^h$	30,000	Daily production capacity (unit)
$Cinvr_{pt}^h$	1	Unitary raw material inventory cost ($ per kg)
$Cinvf_{pt}^h$	0.1	Unitary final product inventory cost ($ per unit)
Cdo_p^h	8	Unitary raw material late-order cost ($ per kg)
$Cpro_{pt}^h$	0.3	Unitary production cost ($ per unit)
Cbl_{pt}^h	3	Unitary backlog cost ($ per unit)
Cvp_p^h	4	Unitary virgin plastic cost ($ per kg)
Cwp_p^h	1	Unitary plastic waste cost ($ per kg)
Ccv_p^h	300	Contract cost for virgin plastics ($)
Ccw_p^h	100	Contract cost for recycled plastics ($)
Cr_p^h	2	Unitary recycling cost ($ per kg)

Table B2
General setting for the rolling LSTM model.

Parameter	Setting	Description
LSTM unit	512	Number of neurons (hidden states) in the LSTM model.
Batch size	16	Number of training samples processed per iteration.
Epochs	70	Number of times the model iterates over the entire dataset during training.
Optimizer	Adam	Optimization algorithm used to minimize the loss function during the training phase.
Learning rate	0.0001	Hyper-parameter used to govern the pace at which the LSTM model updates or learns the values of a parameter estimate.
Stopping criteria	Early stopping	In the present study, by defining patience = 10, the training process will stop if validation RMSE does not improve for 10 consecutive epochs.

Data availability

Data will be made available on request.

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